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19 / 20 submitted

Q1. Void Function (No Return Value)

CODE

```
def welcome():  
    print("Welcome to python Programming")  
welcome()
```

OUTPUT

(no output)

Q2. Void Function (No Return Value)

CODE

```
def table(n):  
    for i in range(1, 11):  
        print(n, "x", i, "=", n*i)  
num = int(input())  
table(num)
```

INPUT

5

OUTPUT

(no output)

Q3. Void Function (No Return Value)

CODE

```
def even_numbers(n):  
    for i in range(1, n+1):  
        if i % 2 == 0:  
            print(i, end=" ")  
n = int(input())  
even_numbers(n)
```

INPUT

10

OUTPUT

(no output)

Q4. Fruitful Function (Returns One Value)

CODE

```
def count_Vowels(s):  
    vowels = "aeiouAEIOU"  
    count = 0  
    for ch in s:  
        if ch in vowels:  
            count += 1  
    return count  
s = input()  
print(count_vowels(s))
```

INPUT

SIMATS

OUTPUT

(no output)

Q5. Fruitful Function (Returns One Value)

CODE

```
def largest_elements(lst):
    largest = lst[0]
    for num in lst:
        if num > largest:
            largest = num
    return largest
lst = list(map(int, input().split()))
print(largest_elements(lst))
```

INPUT

10 25 7 89 34

OUTPUT

(no output)

Q6. Fruitful Function (Returns One Value)

CODE

```
def is_prime(n):
    if n ≤ 1:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True
num = int(input())
if is_prime(num):
    print("Prime")
else:
    print("Not Prime")
```

INPUT

7

OUTPUT

(no output)

Q7. Function Returning Two or More Values

CODE

```
def result(a, b, c):
    t = a + b + c
    avg = t / 3
    if avg ≥ 50:
        g = 'Pass'
    else:
        g = 'Fail'
    return t, avg, g
x, y, z = map(int, input().split())
tot, avg, grade = result(x, y, z)
print(tot, avg, grade)
```

INPUT

70 80 90

OUTPUT

(no output)

Q8. Function Returning Two or More Values**CODE**

```
def circle(r):  
    pi = 3.14  
    area = pi * r * r  
    circum = 2 * pi * r  
    return area, circum  
r = float(input())  
a,c = circle(r)  
print(round(a, 2))  
print(round(c, 2))
```

INPUT

7

OUTPUT

(no output)

Q9. Function Returning Two or More Values**CODE**

```
def calc(n):  
    sq = n * n  
    cu = n * n * n  
    sr = n ** 0.5  
    return sq, cu, sr  
num = int(input())  
s, c, r = calc(num)  
print(s)  
print(c)  
print(round(r, 2))
```

INPUT

4

OUTPUT

(no output)

Q10. Keyword Arguments**CODE**

```
def student(name, age, department):  
    print("Name:", name)  
    print("Age:", age)  
    print("Department:", department)  
student(name="Arun", age=20, department="CSE")
```

OUTPUT

(no output)

Q11. Keyword Arguments**CODE**

```
def employee(empid, name, salary):  
    print("Employee ID:", empid)  
    print("Name:", name)  
    print("Salary:", salary)  
employee(salary=50000, name="Arun", empid=101)
```

OUTPUT

(no output)

Q12. Keyword Arguments

CODE

```
def book(title, author, price):  
    print("Title:", title)  
    print("Author:", author)  
    print("Price:", price)  
book(price=350, title="Python Basics", author="Ravi")
```

OUTPUT

(no output)

Q13. Variable-Length Arguments (*args)

CODE

```
def sum_numbers(*args):  
    total = 0  
    for num in args:  
        total += num  
    return total  
print(sum_numbers(1, 2, 3))  
print(sum_numbers(10, 20, 30, 40))  
print(sum_numbers(5))
```

OUTPUT

(no output)

Q14. Variable-Length Arguments (*args)

CODE

```
def average_marks(*args):  
    if len(args) == 0:  
        return 0  
    total = sum(args)  
    avg = total / len(args)  
    return avg  
print(average_marks(80, 90, 70))  
print(average_marks(45, 60, 75, 90))  
print(average_marks(100))
```

OUTPUT

(no output)

Q15. Variable-Length Arguments (*args)

CODE

```
def print_strings(*args):  
    for s in args:  
        print(s)  
print_strings("Hello", "Python", "SIMATS")  
print_strings("Apple", "Banana", "Orange")
```

OUTPUT

(no output)

Q16. Variable-Length Arguments (*args)

CODE

```
def find_largest(*args):  
    return max(args)  
print(find_largest(10, 25, 5, 40))  
print(find_largest(3, 7, 2))  
print(find_largest(100, 50, 200, 150))
```

OUTPUT

(no output)

Q17. Recursive Functions

CODE

```
def fib(n):  
    if n ≤ 1:  
        return n  
    return fib(n-1) + fib(n-2)  
n = int(input())  
for i in range(n):  
    print(fib(i), end=" ")
```

INPUT

7

OUTPUT

(no output)

Q18. Recursive Functions

CODE

```
def sum_digits(n):  
    if n == 0:  
        return 0  
    return n % 10 + sum_digits(n // 10)  
n = int(input())  
print(sum_digits(n))
```

INPUT

12345

OUTPUT

(no output)

Q19. Recursive Functions

CODE

```
def reverse_string(s):  
    if len(s) == 0:  
        return s  
    return s[-1] + reverse_string(s[:-1])  
s = input()  
print(reverse_string(s))
```

INPUT

SIMATS

OUTPUT

(no output)

Q20. Recursive Functions

— Not Submitted —